

1. Is a V-model (sequential) or incremental model more appropriate for this project? Why?

Conclusion: Incremental Model is more appropriate

Reasons:

a. Complexity and Risk Management

The Omninet project involves multiple subsystems:

- **Kiosk hardware and software** (touchscreen, payment acceptance, browser)
- **Cloud backend** (API Gateway, microservices, databases)
- **AI Support Center** (human agents + AI agents)

Attempting to deliver all these components at once in a strict V-model would be extremely high-risk. An incremental approach allows the team to build and test core functionality first (e.g., basic payment and browsing) and then add advanced features (AI Assistant, content filtering, support center) in later increments.

b. Market Pressure and Release Date

The MRD states: *"The first set of 1,000 Omninet kiosks shall be live, accepting payment and accessing the Internet, as of the financial third quarter."*

This implies a **fixed deadline** and a need to deliver a working product quickly. An incremental model allows the team to release a **minimum viable product (MVP)** that meets the core requirements (payment + Internet access) by the deadline, while continuing to develop and integrate advanced features (AI, support center) in parallel for future updates.

c. Feedback and Adaptation

Public kiosks will receive real-world user feedback. An incremental model allows the team to incorporate that feedback into later increments. For example:

- **Increment 1:** Basic cash and card payments, English only, no AI.
- **Increment 2:** Add more languages and digital wallets.

- **Increment 3:** Integrate AI Assistant and content filtering.

This aligns with the MRD's mention of "*AI will be integrated*" — something that can be added after the basic system is stable.

d. Integration Complexity

The architecture diagram shows multiple layers (Kiosk → Cloud → Support Center). An incremental approach, such as the **backbone integration strategy** previously discussed, allows the team to build and test the communication backbone first, then add layers of functionality. This reduces the risk of integration hell at the end of the project.

e. V-Model Limitations

A strict V-model would require **all requirements to be fully** defined and frozen before development begins. However, the MRD includes some **emerging technologies** (AI agents) and mentions that screen prototypes were "*currently unavailable*" in the original document. This indicates some requirements may evolve, making a purely sequential model impractical.

2. Are maintenance, integration, or verification and validation important for this project? Why?

Conclusion: All three are critically important.

Aspect	Importance	Evidence from MRD
Maintenance	High	Section 3.2.1 Software Updates explicitly describes automatic updates at 2:00 AM for OS patches, drivers, payment rates, virus definitions, and blocked sites. This shows that maintenance is a built-in feature, not an afterthought. Additionally, kiosks in public places require ongoing physical and software maintenance to remain operational.

Aspect	Importance	Evidence from MRD
Integration	High	The system has three major subsystems (Kiosk, Cloud, Support Center) that must work together seamlessly. The MRD mentions: <i>"Call center agents may also push software updates to kiosks"</i> and <i>"Kiosks shall connect to the server farm once per hour to report status."</i> These require robust integration between kiosks, cloud services, and the support center. The architecture diagram (API Gateway, microservices, message queues) further emphasizes integration complexity.
Verification & Validation	High	The MRD contains numerous requirements that must be verified: • Payment processing (must work correctly with cash, cards, digital wallets) • Performance (e.g., 50 KBPS minimum speed) • Security (content filtering, virus protection) • Usability (language support, screen prototypes) • Reliability (session termination, confidentiality) The existence of a <i>"System Requirements Document"</i> and <i>"Screen Prototype Document"</i> (referenced in MRD) shows that validation against user expectations is planned. The inspection report previously conducted also demonstrates the importance of V&V early in the lifecycle.

Why Each is Important:

Maintenance

- Kiosks are deployed in **public places** (airports, malls) and must remain operational 24/7.
- Security threats evolve rapidly; virus definitions and blocked site lists must be updated frequently.
- Payment rates may change over time (inflation, promotions), requiring remote updates.

Integration

- The system is **distributed** — kiosks cannot function without the cloud for payment authorization and AI services.
- The support center needs **real-time data** from kiosks to monitor and assist users.
- Failure in integration could lead to **lost revenue** (payments not processed) or **security breaches**(kiosks not receiving updated threat definitions).

Verification and Validation

- **Verification:** Are we building the system right?
 - Does the payment logic match the decision table?
 - Does the kiosk clear all user data after session end?
 - Does the AI Assistant actually support 20 languages?
 - **Validation:** Are we building the right system?
 - Do users find the Welcome screen inviting?
 - Is the content filter blocking the right material without over-blocking?
 - Are call center agents able to effectively monitor and control kiosks?
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Summary Table

Aspect	Importance	Key Reason
Development Model	Incremental	Complexity, fixed deadline, need for feedback, integration challenges

Aspect	Importance	Key Reason
Maintenance	High	Ongoing updates (software, security, rates) for public kiosks
Integration	High	Three subsystems must work together; real-time communication required
Verification & Validation	High	Payment accuracy, security, usability, and performance must be confirmed

The Omninet project's success depends on **incremental delivery** to manage risk and meet deadlines, combined with strong focus on **maintenance, integration, and V&V** to ensure a reliable, secure, and user-friendly system in public spaces.